

A Prompt-Guided TransUNet Model for Multi-Organ Ultrasound Image Classification and Segmentation

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Abstract. We present Prompt-Guided TransUNet, a TransUNet-based omni-model designed for multi-task ultrasound image analysis. Our approach incorporates task-aware and anatomy-aware prompts into the Vision Transformer encoder, enriching prior knowledge and guiding attention toward clinically relevant structures. To ensure compatibility with pretrained weights, we adopt an adaptive resizing strategy for position embeddings across different grid resolutions. In addition, to further enhance the robustness of our model, we applied a series of diverse pre-processing strategies tailored to different types of images. These strategies were extensively validated through experiments and subsequently integrated into the test-time augmentation process, leading to improved robustness and prediction performance. Experiments on multiple ultrasound datasets show that Prompt-Guided TransUNet achieves a highest score of 82.87 in local testing.

Keywords: Ultrasound Imaging Processing · Orchestration Learning · All-in-One Model

1 Introduction

Ultrasound imaging is a widely used diagnostic tool valued for its real-time capability, safety, portability, and cost-effectiveness, and is essential for examining a wide range of anatomical structures.

Nevertheless, automated ultrasound image analysis remains challenging due to low signal-to-noise ratio, speckle noise, and large variability caused by acquisition settings, operator dependence, and anatomical differences. Existing deep learning models are typically task-specific, trained on limited datasets, and struggle to generalize across organs and scenarios, making the deployment of multiple specialized models inefficient and costly.

Given these challenges, there is a growing need for more robust and adaptable ultrasound image analysis methods that can handle the inherent variability of different ultrasound images while maintaining diagnostic accuracy. To address

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this, prior works like Lin et al. [2, 3] have explored that enriching the model with prior prompts and adopting more effective integration strategies consistently enhance performance across both classification and segmentation tasks. Motivated by these studies, we conducted the following explorations to advance our approach:

- We encode medical priors as position/task/type/nature prompts and, at each transformer block, project them to per-head representations that multiplicatively scale the query vectors, thereby steering attention toward lesion-relevant structures.
- We establish a robust ultrasound preprocessing pipeline . At test time, we apply a lightweight TTA with ensembling, which provides additional accuracy gains of 0.4%.

2 Method

The TransUNet [1] architecture has been extensively proven to demonstrate superior performance in segmentation tasks for medical ultrasound images. Therefore, we enhance the TransUNet framework by incorporating medical prior knowledge from the data in the form of prompts, which are fused into the transformer blocks via the attention mechanism. This approach improves the network’s focus on diverse lesion distributions and characteristics, further unlocking its feature extraction capability for various ultrasound images and enhancing multi-task performance in downstream classification and segmentation tasks. The overall architecture is illustrated in Figure 1.

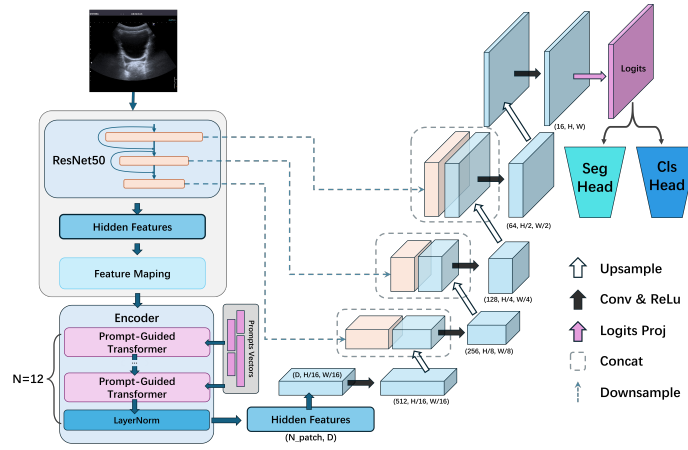


Fig. 1. Overview of Prompt-Guided TransUNet

2.1 Data Preprocessing

Our data preprocessing pipeline integrates a combination of **photometric** and **geometric** transformations, along with **normalization** and **prompt preparation**, to improve model robustness and generalization. Specifically, we apply random gamma correction and contrast adjustment to simulate variations in imaging conditions. For spatial augmentations, we include random rotations ($\pm 30^\circ$), horizontal and vertical flipping, and conditional flipping for **tumor-specific samples** to enhance invariance to anatomical orientations. Additionally, we perform random scaling and cropping or direct resizing to a fixed target size, ensuring consistent input dimensions. All images are subsequently converted into tensors and standardized using ImageNet mean and standard deviation normalization. For testing, a resize-only strategy is used to maintain data consistency without introducing stochastic variations.

2.2 Model Architectures and Multi-task Paradigm

We enhance the standard TransUNet architecture by introducing a prompt-guided attention mechanism into the Vision Transformer decoder, enabling the network to explicitly leverage prior medical knowledge during feature learning. Specifically, four categories of prompts—position, task, type, and nature—are designed to encode heterogeneous prior information about ultrasound imaging. Each prompt is first represented as a one-hot vector, reflecting discrete knowledge such as anatomical region, clinical objective (segmentation or classification), lesion type, or imaging nature. These one-hot vectors are then mapped into the hidden embedding space through learnable linear projections, ensuring compatibility with the dimensionality of the transformer tokens.

Unlike standard token concatenation, this method injects prompts at every Transformer block, allowing the network to continuously refine its representations under the influence of domain priors. To achieve fine-grained modulation, the projected prompt embeddings are reshaped into per-head vectors and combined with the queries in the self-attention module via learnable Feature-wise Linear Modulation (FiLM)-style scaling factors. This design enables each attention head to selectively adjust its sensitivity to specific prompts, thereby controlling how information is aggregated across spatial locations.

Through this mechanism, prompts act as task- and anatomy-aware priors that steer self-attention toward clinically relevant structures, such as organ boundaries or lesion regions, while suppressing irrelevant noise or artifacts. Consequently, the model is capable of adapting its feature extraction process to the requirements of different medical imaging contexts, supporting both segmentation and classification within a unified framework. Furthermore, by disentangling multiple types of prompts, the network can better generalize across heterogeneous datasets, effectively bridging domain gaps that arise from variations in organs, acquisition protocols, and diagnostic tasks.

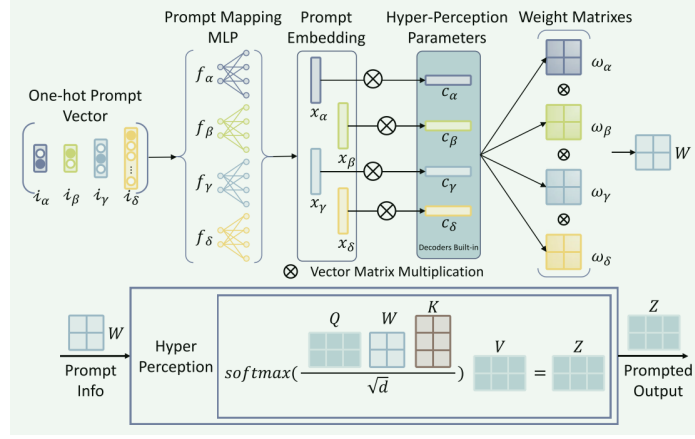


Fig. 2. Prompts Injection Paradigm [2]

2.3 Loss Function

In segmentation tasks, we utilize a combined loss function of Dice loss and cross-entropy loss (CE loss), with distinct weighting coefficients for each ($0.28 \times \text{CE} + 0.72 \times \text{Dice}$). In classification tasks, We observed that certain datasets (e.g., private_breast_luminal) exhibit a pronounced class imbalance; therefore, we adopted a class-weighted cross-entropy loss to compensate for underrepresented categories and stabilize training process.

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log(p_{ic}), \quad (1)$$

where N is the total number of pixels, $y_{ic} \in \{0, 1\}$ indicates the ground-truth label, and $p_{ic} = \text{softmax}(z_{ic})$ is the predicted probability.

$$\text{Dice}_c = \frac{2 \sum_{i=1}^N p_{ic} y_{ic} + \epsilon}{\sum_{i=1}^N p_{ic} + \sum_{i=1}^N y_{ic} + \epsilon}, \quad (2)$$

$$\mathcal{L}_{Dice} = 1 - \frac{1}{C} \sum_{c=1}^C \text{Dice}_c, \quad (3)$$

where ϵ is a small constant for numerical stability.

3 Experimental

3.1 Implementation Details and Experimental Settings

All experiments were conducted using a distributed training paradigm with `torch.distributed` (DDP, NCCL backend), where each GPU was assigned an

individual process. The proposed framework jointly optimized segmentation and classification tasks with carefully designed sampling, augmentation, and optimization strategies. To address the imbalance across datasets, we employed a customized `WeightedRandomSamplerDDP`, ensuring fair contributions from each dataset during training.

The optimizer was AdamW with an initial learning rate $\eta_0 = 0.0001$, weight decay 0.01, and momentum coefficients $(\beta_1, \beta_2) = (0.9, 0.999)$. A polynomial-like decay schedule was adopted: for segmentation, the effective learning rate followed $\eta = \eta_0 \times (\text{progress})^{0.9}$ but remained lower-bounded at $(0.2)^{0.9}$, whereas for classification the threshold was $(0.3)^{0.9}$.

Our experiments reveal that joint training of classification and segmentation leads to mutual interference between the two tasks. Therefore, we opted to decouple the two tasks. The training protocol comprises two phases: initially performing exclusive segmentation training for 130 epochs, followed by freezing all non-classifier parameters and proceeding with dedicated classification training for 360 epochs to derive the ultimate model weights.

3.2 Results

During local evaluation, we adopted the Dice coefficient and accuracy as the primary metrics for segmentation and classification, respectively.

Table 1. Comparison of different backbone designs on private datasets. Dice coefficient is reported for segmentation tasks, and Accuracy for classification tasks. The best results are highlighted in bold.

Segmentation (Dice)			
Dataset	Baseline [2]	MultiDecoders	PG-TransUNet (ours)
Thyroid	0.5042	0.5932	0.6747
Kidney	0.8235	0.8469	0.9204
Fetal_Head	0.8785	0.9283	0.9220
Cardiac	0.8474	0.8634	0.8646
Breast_luminal	0.7060	0.8189	0.8617
Breast	0.8816	0.8346	0.8693
Mean	0.7735	0.8142	0.8521
Classification (Accuracy)			
Dataset	Baseline [2]	MultiDecoders	PG-TransUNet (ours)
Liver	0.8387	0.6774	0.8182
Breast_luminal	0.8732	0.8169	0.7083
Breast	0.7778	0.7111	0.7333
Appendix	0.7619	0.8571	1.0000
Mean	0.8129	0.7656	0.8150

In order to evaluate the effectiveness of adopting TransUNet as the backbone, we compared it against [2] with a single encoder-decoder structure, as well as an

improved variant with a single encoder and multiple decoders (MultiDecoders). In the MultiDecoders design, each organ-specific dataset was assigned a dedicated decoder to mitigate interference across tasks. Although this approach provided some performance gains compared to the baseline, the improvements were still limited. By contrast, the Prompt-Guided TransUNet (PG-TransUNet) model consistently outperformed both alternatives across most tasks, achieving the best overall segmentation and classification accuracy. The quantitative results are summarized in Table 1.

Table 2 reports the results on private datasets before and after applying test-time augmentation (TTA). We observe that TTA brings consistent improvements in overall segmentation (Mean Dice: 0.8484 $\rightarrow$ 0.8521) and classification (Mean Acc: 0.8087 $\rightarrow$ 0.8150) performance, with notable gains on thyroid segmentation and breast classification.

Table 2. Performance on private datasets before and after applying test-time augmentation (TTA). Dice coefficient is reported for segmentation tasks, and Accuracy for classification tasks.

Segmentation (Dice)			Classification (Accuracy)		
Dataset	w/o TTA	w/ TTA	Dataset	w/o TTA	w/ TTA
Thyroid	0.6498	0.6747	Liver	0.8182	0.8182
Kidney	0.9206	0.9204	Breast_luminal	0.7500	0.7083
Fetal_Head	0.9223	0.9220	Breast	0.6667	0.7333
Cardiac	0.8647	0.8646	Appendix	1.0000	1.0000
Breast_luminal	0.8527	0.8617			
Breast	0.8803	0.8693			
Mean	0.8484	0.8521	Mean	0.8087	0.8150

4 Discussion

From Table 1, we observe that the baseline single encoder–decoder struggles with heterogeneous ultrasound data, showing poor segmentation Dice for thyroid and breast_luminal. The MultiDecoders design improves segmentation (mean Dice 0.8142 vs. 0.7735) by task-specific decoders, but reduces classification accuracy (0.7656 vs. 0.8129) due to limited feature sharing. In contrast, the TransUNet backbone achieves the best balance, with the highest Dice (0.8521) and competitive accuracy (0.8150), highlighting the advantage of Transformer-based attention for modeling diverse anatomical structures.

From the comparison in Table 2, TTA is shown to provide consistent benefits. For segmentation, the mean Dice improves from 0.8484 to 0.8521, with notable gains in *Thyroid* (0.6498 $\rightarrow$ 0.6747) and *Breast_luminal* (0.8527 $\rightarrow$ 0.8617), suggesting that multi-view predictions mitigate local noise and enhance robustness. For classification, the mean accuracy increases from 0.8087 to 0.8150, with clear improvement in *Breast* (0.6667 $\rightarrow$ 0.7333), though a slight drop is observed in

Breast_luminal. Overall, these results indicate that TTA yields stable or modest gains in both segmentation and classification, highlighting its effectiveness in handling noisy ultrasound images.

5 Conclusion

In this work, we demonstrated that integrating prompt-guided priors into the TransUNet backbone significantly enhances the model’s ability to capture clinically relevant structures across heterogeneous ultrasound datasets. The enriched prompts effectively guide the attention mechanism, improving both segmentation and classification performance. In addition, we established a tailored preprocessing pipeline for ultrasound data, which standardizes input quality and further boosts model robustness. Finally, incorporating test-time augmentation (TTA) provides additional gains by mitigating noise sensitivity and stabilizing predictions. Together, these strategies yield a strong and generalizable framework for multi-organ ultrasound analysis.

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